

Gold-Free Faithfulness Metrics for NL-to-FOL Translation: Executive Summary

AI Inventor

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1 Problem

Evaluating whether a candidate first-order logic (FOL) formula faithfully captures a natural language sentence is an open problem. Gold FOL annotations are expensive, non-unique, and unreliable (approximately 42% error rate in FOLIO and MALLS). This investigation searched for gold-free metrics that predict faithfulness given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge.

2 Primary Finding: Cross-Family Consensus

The strongest gold-free signal is **cross-family solver consensus** (`c_score_align`): collect translations of the same sentence from multiple independent LLM families, check z3-equivalence between the candidate and each peer, and score by agreement fraction. The mechanism is that errors scatter across z3-equivalence classes while correct translations cluster.

Table 1: Headline results: `c_score_align` vs baselines across evaluation sets.

Metric	Screen (AUROC)	Dev set E (strat AUROC)	R_COMP SIG (AUROC)
<code>c_score_align</code> (V0)	0.866	0.742	0.954
PEER+TEXT fusion	0.870	0.790 (pooled)	—
Cheap LLM judge (disg.)	0.777	0.641	0.587
FOL-Triage fused	0.759	0.770 (R_ADJ)	—
Round-trip NLI	0.710	—	—
Pilot structural metrics	0.50–0.53	—	—

Key quantitative results:

- On development set E, `c_score_align` beats the best cheap judge by +0.101 [+0.053, +0.145] stratified AUROC (T1 CONFIRMED, iteration 3).
- Three peer families suffice for 97% of full-pool performance ($k_{95} = 3$; by word-count tercile $k_{95} = 2/3/5$).
- On R_COMP SIG (controlled vocabulary), `c_score_align` achieves 0.954 within-template AUROC with zero error endorsement.
- Adding `c_score_align` to the best baseline combination (S4_full) yields +0.039 [+0.021, +0.057] strat AUROC ($p_{95} = 0.009$).

- Per-item cost is \$0.0015/sentence (peer generation) plus 2.5×10^{-7} /item (z3 calls).

3 The Rename Tradeoff: Central Open Problem

V0 has RENAME_NONCE false-alarm rate 0.765 on development E and 1.0 on E2: any renaming of predicates breaks the consensus signal, because two formulas with different predicate names are never z3-equivalent. This is fundamental to the z3-equivalence mechanism.

Two approaches to solve this failed:

- **Candidate-Signature Consensus (CSC, iteration 4):** Feed peers the candidate’s own predicates. Error endorsement rose from 0.173 to 0.459 (anchoring effect). Strat AUROC dropped to 0.614. DEAD END.
- **Gloss-Gated agreement (GG, iteration 5):** Filter predicate mappings via natural-language glosses. Reduced MEANING_RENAME endorsement from 0.502 to 0.224, but strat AUROC dropped from 0.742 to 0.696 (gloss gate rejects legitimate mappings; $d = 0.629$). DEAD END.

4 What Worked and What Did Not

Table 2: Dead ends across all iterations.

Dead end	Iter.	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware check	1	AUROC 0.557 (below bag-of-words)
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c.nf)	2–3	SWAP recall 0.512
CSC (candidate vocabulary)	4	Strat AUROC 0.614, anchoring $e = 0.459$
GG (gloss-gated)	5	Strat AUROC 0.696 vs V0 0.742

No consensus variant V1–V5 improved on V0. The best candidate (V4, two-channel logistic) gained only +0.0088 LONG-strat, well below the +0.015 selection threshold. V0 stands as the shipped metric.

5 Limitations

1. **No confirmation completed.** All three iteration-5 confirmation arms (E2-A, E2-B, R_COMP FREE) were budget-stopped before reaching testability. No confirmatory hypothesis is confirmed or refuted.
2. **Consensus degrades with sentence length.** NET +0.401 on development E, driven by correct-row divergence d . Long sentences need more peers and the metric is less reliable precisely where it matters most.
3. **Domain transfer untested.** All evaluation is on FOLIO and MALLS-derived data.

4. **Label noise.** Panel majority accuracy on expert-curated errors is 0.727; R_COMP FREE audit shows 13–28% label contamination.
5. **Rename tradeoff unsolved.** No variant achieves both low rename FA and high error recall.

6 Run Trajectory

Table 3: Run trajectory: one row per iteration.

Iter.	Hypothesis title	Move	Review	Executed	Ledger spend
1	Peer agreement plus text checks vs judges	deepen	3	yes	\$17.37
2	Peer agreement vs judges on long sentences	deepen	3	yes	\$34.76
3	Peer agreement in the candidate’s own words	deepen	3	yes	\$84.75
4	Cross-model agreement, tested on fresh data	deepen	3	yes	\$159.80
5	Cross-model agreement, unconfirmed on fresh data	declare	3	yes	\$315.97

The full report (52 pages, 111 tables, 8 figures) provides detailed results for all 20 artifacts across 5 iterations.